

DeepThinkTrader v2.0 – Complete System Guide

Generated: March 20, 2026 | **Repository:** github.com/RebelHawk-TK/DeepThinkTrader

CRITICAL WARNING: This is a paper trading bot. Never use real money until backtested for 3+ months. Trading bots can lose 100% of capital. This project is educational and experimental.

1. Architecture Overview

Pipeline: Scanner → Research Agent → DeepThink Analysis (Rule-Based + Multi-Edge) → Risk Gate (13 checks) → Execution Agent → Alpaca Paper Trading

Component	Technology	Purpose
Scanner Agent	Alpaca Screener API	3-stage funnel: discovery, pre-screen, scoring — main + penny (\$1-\$5)
Research Agent	Alpaca + Twelve Data + NewsAPI + Yahoo Finance	Gathers technicals, fundamentals, news sentiment, analyst ratings, insider trades, earnings data
DeepThink Agent	Rule-based scoring + multi-edge validation	Conviction scoring, scenario modeling, 3-edge confirmation (Fund/Tech/Sentiment)
Risk Manager	Kelly criterion + portfolio guards	13 pre-trade checks: Kelly sizing, drawdown halt, circuit breaker, liquidity, earnings
Execution Agent	Alpaca Paper Trading API	Market/limit orders, trailing stops, partial scale-out, time stops, slippage tracking
Dashboard	Streamlit + Plotly	Real-time portfolio monitoring, trade history, strategy health
Database	SQLite	Trade logging, research reports, trailing stops, partial exits, API request IDs

2. Data Sources

Source	Data Provided	API Key Required	Free Tier
Alpaca Markets	Price, volume, SMA, RSI, trade execution, portfolio history, SPY snapshots	Yes (paper account)	Unlimited

Twelve Data (RapidAPI)	MACD, Bollinger Bands, EMA 9/21, Stochastic, ADX, ATR	Yes (RapidAPI key)	800 req/day
NewsAPI	News headlines + VADER sentiment scoring	Yes	500 req/day
Yahoo Finance	P/E, ROE, growth, debt/equity, earnings dates, insider trades, institutional holdings	No	Unlimited
Reddit (PRAW)	Sentiment from r/wallstreetbets, r/stocks, r/investing, r/pennystocks	Yes (optional)	60 req/min

3. Risk-First Framework (v2.0)

Philosophy: Every trade must pass through a rigorous, emotion-proof checklist before execution. No trade executes on conviction alone — it requires multi-edge confirmation and portfolio-level safety.

3.1 Position Sizing: Fractional Kelly

Condition	Method	Formula
20+ closed trades	Fractional Kelly	$f^* = (p - q) / b \times 0.5$ (half-Kelly)
< 20 trades	Fixed Risk	1% of equity per trade

Where p = win rate, $q = 1 - p$, b = avg win / avg loss (payoff ratio). Capped at MAX_POSITION_PCT.

3.2 Pre-Trade Checks (13 gates)

#	Check	Action if Failed
1	Conviction meets threshold	HOLD
2	Risk within limit	BLOCKED
3	Reward:Risk ratio acceptable	BLOCKED
4	Daily loss limit OK	BLOCKED for rest of day
5	Open position count OK	BLOCKED
6	No duplicate position	BLOCKED
7	Market hours	BLOCKED
8	Drawdown from peak < 8%	BLOCKED (all new entries)
9	Risk of ruin < 1%	BLOCKED
10	Liquidity: shares < ADV/5	Auto-reduce or BLOCKED
11	Multi-edge: 2/3 edges firing	HOLD
12	Market health: SPY not down > 2%	BLOCKED (longs only)
13	No earnings within 2 days	Auto-close or BLOCKED

3.3 Volatility Adjustment

If current ATR > 3× median ATR (50-day), risk percentage is automatically cut in half.

3.4 Revenge Trading Detector

If last 3 trades are all losses, all new entries are blocked until a winning trade occurs.

4. Multi-Edge Validation (v2.0)

Rule: A trade requires at least **2 of 3 independent edges** to execute. Conviction score alone is not enough.

4.1 Fundamental Edge

Must pass at least 3 of 4 criteria:

Criterion	Threshold
P/E ratio	< 20 or forward P/E improving
Return on Equity	> 15%
Earnings/Revenue Growth	> 10% YoY
Debt/Equity	< 100

4.2 Technical Edge

Must pass at least 2 of 3 criteria:

Criterion	Threshold
Price vs 200-day SMA	Above (for longs)
RSI(14)	< 45 (neutral-bullish) or < 30 (oversold)
Volume breakout	> 1.5× 20-day average

4.3 Sentiment/Regime Edge

Must pass at least 2 of 3 criteria:

Criterion	Threshold
Combined catalyst score	> 0.2
News sentiment	> 2/10 (bullish)
Reddit sentiment	> 0.3 (bullish)

5. Exit Management (v2.0)

5.1 Exit Check Frequency

Check Type	Frequency	What It Does
Fast exit check	Every 5 minutes	Price checks only — SL/TP/trailing/time stops
Full cycle	Every 60 minutes	Full scan + research + analysis + execution

5.2 Trailing Stops

Parameter	Main Portfolio	Penny Portfolio
Activation	2% profit	2% profit
Trail distance	1.5%	3.0%

Once activated, trailing stop replaces static stop-loss. Static take-profit still applies.

5.3 Partial Scale-Out

Profit Level	Action	Remaining
1R (1× risk amount)	Sell 33% of position	67%
2R (2× risk amount)	Sell another 33%	34%
3R+ or trailing stop	Remaining 34% rides with trail	0%

5.4 Time Stop

If a position shows < 2% movement after 15 trading days, it is automatically closed to free capital.

5.5 Earnings Auto-Exit

If earnings are within 2 trading days, position is automatically closed (configurable: close or tighten SL 50%).

6. Smart Order Execution (v2.0)

6.1 Order Types

Portfolio	Order Type	Details
Main	Market order	Immediate fill, slippage tracked
Penny	Limit order	Current price + 0.5% slippage buffer, day TIF, auto-cancel after 30 min

6.2 Slippage Tracking

After fill, actual price is compared to expected. Deviations > 0.3% are logged as slippage alerts.

6.3 Pre-Trade Transparency

Every trade logs a plain-English summary before execution:

```
BUY 120 shares of XYZ @ $4.50
Risk: $54 (0.1% of portfolio) | Edges: 2/3
R:R = 2.4:1 | Stop: $4.05 | Target: $5.58
Portfolio: penny | ADV: 500,000
```

7. Market Circuit Breaker

Trigger	Action
SPY down > 2% intraday	Block all new LONG entries (shorts still OK)
30-day drawdown > 8%	Block ALL new entries (not exits)
Negative expectancy	Block trades (risk of ruin too high)

8. Post-Trade Learning (v2.0)

8.1 Strategy Health Metrics

Metric	Description
Win Rate (30d)	% of trades that were profitable
Expectancy	(win_rate × avg_win) - (loss_rate × avg_loss)
Profit Factor	Gross wins / gross losses

Payoff Ratio	Avg win / avg loss
WR vs Baseline	30-day win rate vs 90-day baseline

8.2 Kelly Feedback Loop

Win rate and payoff ratio from closed trades feed directly into position sizing. As strategy performs better, Kelly sizes up. As it degrades, sizing shrinks automatically.

8.3 Weekly Health Check (Monday)

If 30-day win rate drops > 15% from 90-day baseline, a degradation warning is logged.

9. Configuration

9.1 Trade Modes

Parameter	Safe	Normal	Aggressive
Max Risk/Trade	1%	2%	3%
Daily Loss Limit	3%	5%	8%
Min Conviction	9.0	7.5	6.0
Min R:R Ratio	3:1	2:1	1.5:1
Max Position %	5%	10%	15%
Max Positions	5	10	15

9.2 Risk-First Parameters (v2.0)

```
# Risk Gate
RISK_PCT_PER_TRADE=0.01      # 1% default (Kelly fallback)
MAX_DRAWDOWN_HALT_PCT=0.08   # 8% drawdown halts trading
VOLATILITY_ATR_MULTIPLIER=3.0 # ATR > 3x median = cut risk 50%
MIN_ADV_RATIO=5               # Shares must be < ADV/5
KELLY_SAFETY_MULTIPLIER=0.5   # Half-Kelly
MAX_RISK_OF_RUIN_PCT=0.01     # Block if RoR > 1%

# Exit Management
EXIT_CHECK_INTERVAL_MINUTES=5
TRAILING_STOP_ACTIVATION_PCT=2.0
TRAILING_STOP_DISTANCE_PCT=1.5
PENNY_TRAILING_STOP_DISTANCE_PCT=3.0
SCALE_OUT_ENABLED=true
SCALE_OUT_LEVELS=1.0,2.0      # R-multiples
TIME_STOP_DAYS=15

# Edge Validation
MIN_EDGES_REQUIRED=2          # 2 of 3 edges must fire

# Smart Orders
PENNY_LIMIT_SLIPPAGE_PCT=0.5
MAX_SLIPPAGE_PCT=0.3

# Market Awareness
CIRCUIT_BREAKER_SPY_DROP_PCT=-2.0
```

```
EARNINGS_EXIT_DAYS=2
EARNINGS_EXIT_MODE=close          # or "tighten"
```

10. Quick Start

```
git clone https://github.com/RebelHawk-TK/DeepThinkTrader.git
cd DeepThinkTrader
cp .env.template .env           # Fill in your API keys
pip install -r requirements.txt
python main.py once              # Test single cycle
python main.py scan              # Test scanner
python main.py penny scan       # Test penny scanner
python main.py ticker NVDA      # Test single ticker with edge validation
streamlit run dashboard.py      # Launch dashboard
./run.sh                        # Start bot + dashboard
./stop.sh                       # Stop everything
```

11. Project Structure

```
StockTrader/
├── main.py                # Orchestrator (scheduled loop + 5-min exit checks)
├── config.py              # Environment config (3 modes + 20 risk params)
├── dashboard.py           # Streamlit monitoring + strategy health
├── run.sh / stop.sh / status.sh # Control scripts
├── agents/
│   ├── deepthink_agent.py # Rule-based analysis + multi-edge validation
│   ├── execution_agent.py  # Alpaca trades + trailing stops + scale-out
│   ├── research_agent.py   # Multi-source data gathering
│   └── scanner_agent.py    # 3-stage discovery (main + penny)
├── utils/
│   ├── alpaca_data.py      # Alpaca market data + X-Request-ID
│   ├── database.py         # SQLite (trades, trailing stops, partial exits)
│   ├── risk_manager.py     # Kelly sizing + 13 pre-trade checks
│   ├── twelve_data.py      # MACD, Bollinger, EMA, Stochastic, ADX, ATR
│   └── yahoo_fundamentals.py # Fundamentals, earnings calendar, edge scoring
├── .env.template          # API key template
├── requirements.txt        # Python dependencies
└── docs/                  # Documentation + PDF guides
```

DeepThinkTrader v2.0 — Risk-First Framework — Paper Trading Mode Only — github.com/RebelHawk-TK/DeepThinkTrader

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